Post-Quantum Cryptography Migration

Part 10 - Building & Shipping Software: Containers, Orchestration, Service Mesh

Compiled by the Spaced Research Ledger

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Cloud-native infrastructure got much of its post-quantum protection by accident. Kubernetes, containerd, and their neighbors are written in Go, and when Go turned hybrid key exchange on by default, whole clusters started negotiating it with no announcement and no planning.

This report covers that world in three sections. Section 1 covers the container runtimes, Section 2 the Kubernetes certificate machinery, and Section 3 the service meshes and workload identity.

The pattern to understand is transport versus certificates. Every connection between cluster components can already run hybrid post-quantum key exchange. But every certificate those components present is still classical, and the tools that issue them, from cert-manager to SPIRE, have open issues rather than shipped support.

The meshes are the exception that moved deliberately. Istio ships a named post-quantum compliance policy, and Linkerd went further, making ML-KEM the default for all meshed traffic. When infrastructure inherits good defaults, the remaining work is the part nobody inherits: the signatures.

1. Container Runtimes

Container runtimes are the engines that actually run containers: Docker, containerd, and their enterprise derivatives. Their cryptography secures the control connections between components and the pulls from registries. Most of it comes straight from the Go standard library, which is the whole story of their post-quantum status.

Recent Developments

The enterprise edge is moving deliberately. Mirantis is adding hybrid post-quantum key exchange to its container runtime's internal mTLS, automatic, certificate-preserving, and FIPS-compatible [1]. The measured cost is under half a millisecond and two kilobytes per handshake [1]. containerd needs no such project: built on current Go, it negotiates the hybrid group by default [2].

Current State

The one deliberate decision on record is a retirement. Docker's Notary v1 service behind Docker Content Trust shuts down in December 2026, with publishers directed to Sigstore or Notation rather than any post-quantum migration of the old system [3].

2. Kubernetes PKI

Kubernetes runs on certificates: every node, controller, and API request authenticates with X.509. The transport between components inherited post-quantum key exchange from Go, but the certificate machinery is its own system. This section covers a cluster's split personality, quantum-safe on the wire and classical in every identity it issues.

Recent Developments

The certificate tooling is at the planning stage. cert-manager opened a tracking issue in June to plan ML-DSA certificate support once Go 1.27 ships the algorithm in its certificate library [4].

Current State

The transport story happened without a plan. Kubernetes components are Go programs, and Go's hybrid group became default in Go 1.24 after a draft period [5]. Kubernetes v1.33 moved to that toolchain without overriding preferences, so its components picked up hybrid post-quantum TLS with no enhancement proposal and no announcement [6]. The kubelet and the rest of the control plane gained it the same way [7][6].

The certificates stayed behind. cert-manager's API offers only RSA, ECDSA, and Ed25519 [8]. Kubelet certificate issuance documents no post-quantum signing option at all, so the coverage stops at the handshake [9]. etcd lagged even on transport, called out as the one control-plane component without support while it sat on an older Go [7]. Its releases have since moved through Go 1.24 and 1.25 builds [10][11].

One operational trap deserves attention. The post-quantum default degrades silently. A component built on older Go negotiating with a newer one loses the shared group, falling back to classical key exchange without error [6]. Mixed-version clusters get no warning that their protection varies by connection.

3. Service Mesh & Workload Identity

A service mesh encrypts the traffic between microservices automatically, and workload-identity systems like SPIFFE hand each service its certificate. Together they are the inside-the-cluster PKI. This section covers the two meshes that shipped deliberate post-quantum support, and the identity layer where signatures remain an open request.

Recent Developments

The identity layer is inching forward. SPIRE 1.15.2 added two post-quantum curves and extended its post-quantum policy to the bundle endpoint and metrics server [12][12]. Consul, by contrast, has an open request just asking for the algorithms and clearer documentation, despite its Go foundation carrying them since 1.24 [13].

Current State

Istio made post-quantum a named policy. Version 1.27 added a compliance setting enforcing TLS 1.3 with the hybrid key exchange [14], covering proxy-to-proxy traffic, gateways, and the control-plane channel [15]. Red Hat's mesh, built on Istio 1.28, carries the support into its 3.3 release [16], after documenting a custom-build stopgap for earlier versions [15]. An open issue asks to extend coverage deeper into mesh-internal traffic [17]. The data-plane caveat lives downstream: Consul's mesh runs on vendored Envoy, and upstream Envoy's post-quantum issue has gone stale [18].

Linkerd simply changed the default. Version 2.19 swapped its proxy's cryptographic module, added the stronger cipher suites, and made ML-KEM-768 the default key exchange for all meshed-pod traffic [19]. It exports the negotiated algorithms as metrics, with the FIPS-validated enterprise build offered as a distinct path [20].

SPIFFE and SPIRE hold the certificate gap. An open backlog issue asks what SPIRE needs to issue ML-DSA-signed identities and run ML-KEM connections, indicating none of it was implemented [21]. The shipped work is policy-level, updating the key-exchange requirement to the standardized group [22]. Academic designs for a hybrid post-quantum SPIFFE exist as proposals only [23].

Unresolved Conflicts

One outside account oversells SPIRE. A blog post describes SPIRE's migration to ML-DSA as proceeding smoothly thanks to its plugin structure [24]. That directly contradicts the project's own open, backlog-tagged issue asking what such a migration would even require [21]. The project issue is the primary source. A SPIRE release note shipping ML-DSA identities would settle it.

About This Report

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